Java Enterprise Application Development

Lecture 0
Introduction to the Course

Dr. Fan Hongfei

16 September 2026

Learning Objectives

- Upon successful completion of the course, students will be able to:
  1. Apply Java programming and object-oriented principles to develop robust and maintainable applications.
  2. Develop enterprise applications using web technologies, Spring, data persistence, messaging, and transaction management.
  3. Design and implement enterprise software architectures, integrate AI technologies, and apply appropriate testing and engineering practices.
  4. Analyze requirements, solve technical problems, and make informed design and implementation decisions.
  5. Plan and manage collaborative software projects, work effectively in teams, and communicate technical solutions professionally.
  6. Demonstrate professional engineering practice through quality software development, documentation, presentation, and responsible use of AI tools.

Programs

Week	Date	Program
1	16 Sep 2026	Course Introduction; Java Fundamentals
2	23 Sep 2026	OOP with Java: Classes and Objects, Inheritance and Interfaces
3	30 Sep 2026	Java Core Advanced Features: Exceptions, Generics, Collections, Modern Syntax, Annotations, Reflection; Unit Testing
4	7 Oct 2026	AI-Assisted Java Development: Requirements and Design, Implementation and Debugging, Testing and Review
5	14 Oct 2026	AI-Powered Java Applications: LLM Integration, Retrieval-Augmented Generation, Agent Development
6	21 Oct 2026	Programming Examination
7	28 Oct 2026	Assignment A Presentation (I)
8	4 Nov 2026	Assignment A Presentation (II)
9	11 Nov 2026	Assignment A Presentation (III)
10	18 Nov 2026	Servlet Programming: Core Servlet Mechanisms, MVC Architecture and View Evolution, Session State Management
11	25 Nov 2026	Spring: IoC Container and Dependency Injection, Spring Boot, Spring Web MVC, Aspect-Oriented Programming
12	2 Dec 2026	Data Persistence: JDBC, MyBatis, JPA and Hibernate, Spring Data
13	9 Dec 2026	JMS and JTA: Asynchronous Messaging and Distributed Transaction Coordination
14	16 Dec 2026	Assignment B Presentation (I)
15	23 Dec 2026	Assignment B Presentation (II)
16	30 Dec 2026	Assignment B Presentation (III)

Assessment Components

a) Attendance and participation (10%)

b) Programming examination (20%)

c) Assignment A (20%), team-based

- A software project that deeply involves fundamental Java programming
- Source code, documents, and presentation

d) Assignment B (50%), team-based

- A software project that deeply involves enterprise Java development
- Source code, documents, and presentation

- *For Assignments A and B, each team contains at most 4 members*